CHAPTER 8

Fitting and Delivering the Finished Dentures

1 Finishing dentures using digital manufacturing system

[1] Advantage of digital denture technology

1) Accurate reproduction of anatomy
2) Fewer patient visits
3) Shorter technical working time required for dental technicians and denturists.

Digital denture manufacturing system is separated into printing and milling processes.

Printed dentures are post-cured and very useful for immediate and temporary dentures but not final dentures at present because some of the materials are not biocompatible.

Milled dentures have high accuracy and strength, and are hygienic resins due to a pre-cured system.

Ivotion Digital Denture System

Ivoclar Ivotion. Digital Denture System is the wide range of options.
1st milling step : Oversized milling process of the denture base and teeth.
2nd milling step : Fine milling denture base.
This method is highly recommendable for everyone in order to reduce the working time for occlusal adjustment and finally improve the quality of the final product with SEMCD.



Fig 8-1 To avoid the bonding gap between arches and rough milling base, the 2nd smooth milling will be carried out.

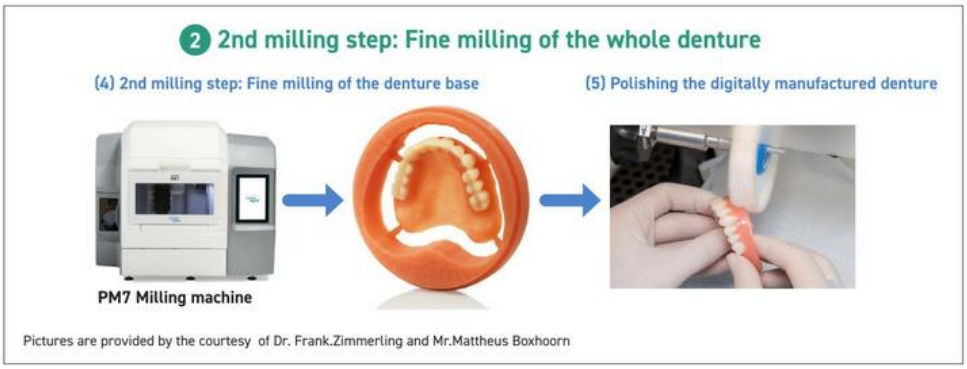


Fig 8-2 The 2nd fine milling process can prevent the teeth from lifting by the 1st rough milling step having the bonding gap.



Fig 8-3 Uneven occlusal contacts can be seen caused by bonding the arches to the denture base with Ivobase CAD Bond, resulting in the necessity of occlusal adjustment.

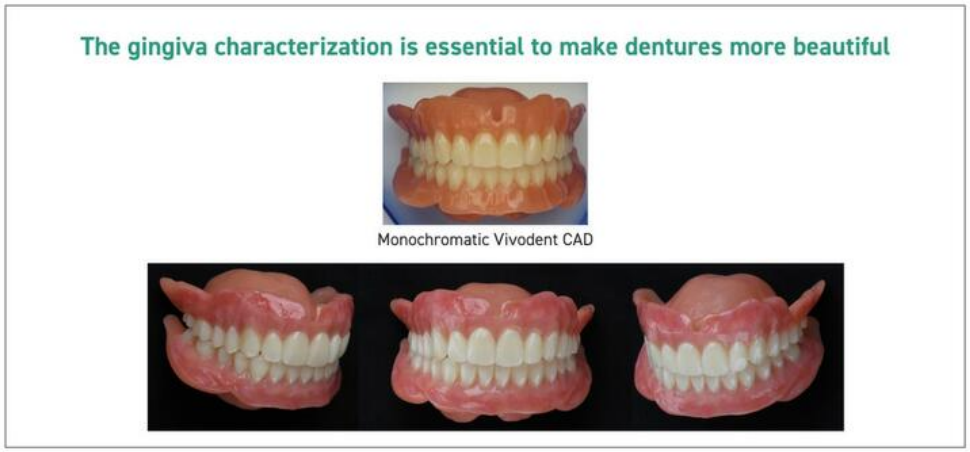


Fig 8-4 Curving cervical areas and gingiva characterization with SR-NEXCO by Mr.Kenji Iwaki.

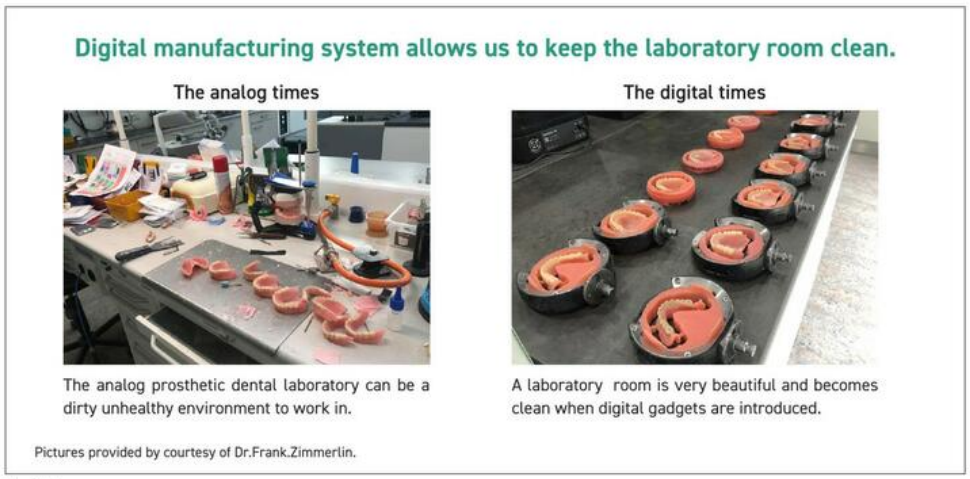


Fig 8-5

2 Fitting the finished digital dentures

[1] PIP (Pressure Indicator Paste) test (Figs 8-6~8)

Following intraoral occlusal adjustments, the oil-base Pressure Indicator Paste (PIP) is applied uniformly over the fitting surface of the denture base. The patient is asked to bite on a cotton roll freely. Complete dentures borne by ridge mucosa constantly move as they function in the mouth. Pressure areas where the paste gets thinner need to be adjusted. Even if the mid-palatal and incisive papilla regions of the maxillary denture are adjusted , suction force will not affected as long as the entire borders are sealed. By the same token, grinding of the fitting surface of the mandibular denture base will not reduce suction force unless the border seal is inadvertently broken.

Procedure



Fig 8-6 Apply the oil-based PIP paste to the interior surface of the denture base evenly.



Fig 8-7 Pressure areas where the PIP paste becomes thinner are marked with a pen and grinded.



Fig 8-8a~c The patient is instructed to bite on a cotton roll freely.

Short Summary

The amount of occlusal clearance between the maxillary and mandibular buccal cusps in lingualized occlusion should be adjusted according to the dietary habit in each country to enrich the patient's diet and increase the joy of eating. For example, Japanese patients who eat more grains, pickles and fish than meat should be given a smaller clearance.

Make sure that the BTC point is formed with the side of the tongue in intimate contact with the buccal mucosa above the retromolar pad on each side.



Fig 8-9 Make sure that the BTC point is formed with the side of the tongue in intimate contact with the buccal mucosa above the retromolar pad.

The form of the borders of the suction-effective maxillary and mandibular digital dentures

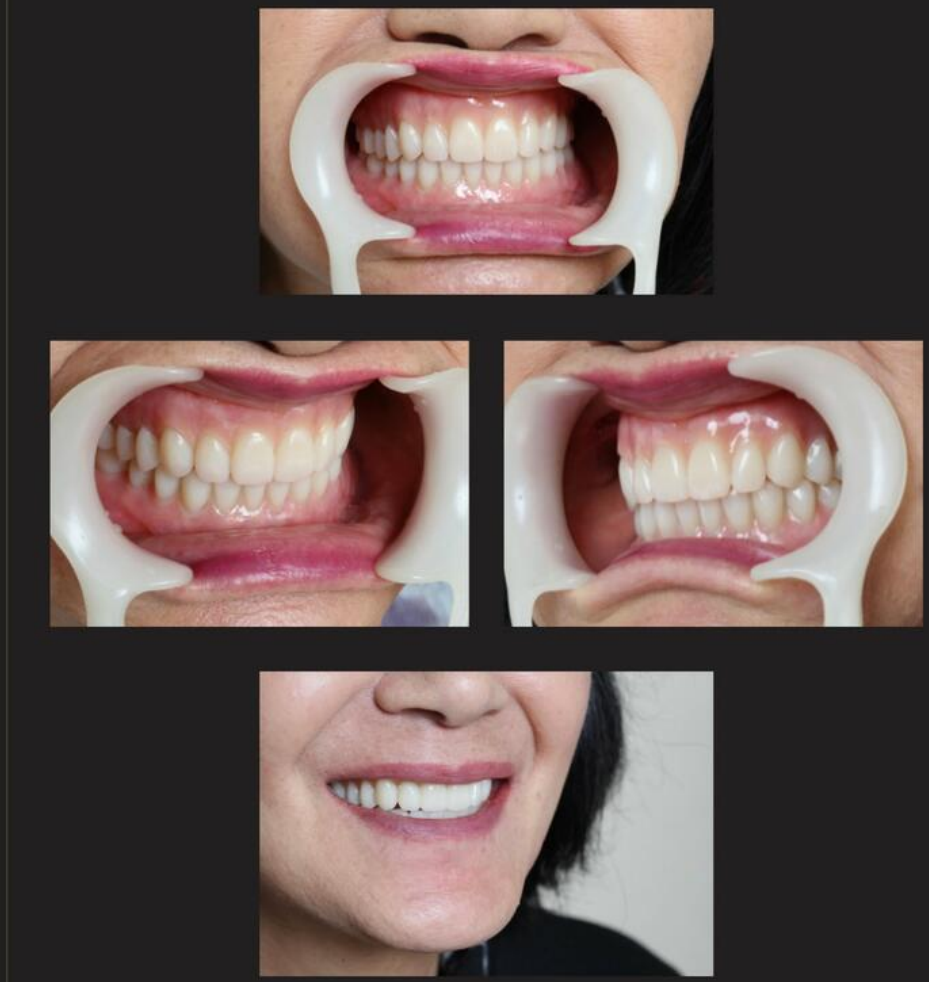


Fig 8-10

Characteristics of the suction-effective mandibular complete digital denture

Retromolar pad is adequately covered.

Someya's sinew string is avoided.

Denture teeth are arranged in the middle of the alveolar ridge to allow close approximation of the cheeks and tongue to the polished surfaces.

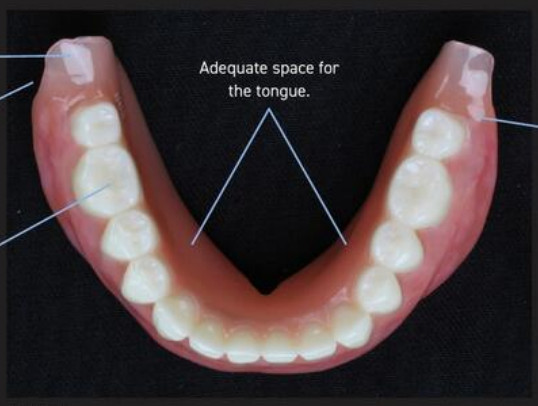


Fig 8-11

The denture border in the retromylohyoid fossa region is thin and extended down approx. 3 mm to complete the compensatory closure.

Passamonti's notch is provided(Deflection point of "S "curve).

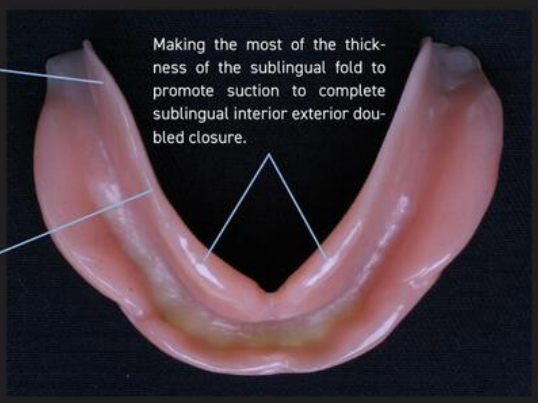


Fig 8-12

The mandibular labial polished surface is made concave from lateral incisor to lateral incisor for lip support.



Fig 8-13 Morphological characteristics of the mandibular denture fabricated based on a good understanding of the concept of denture suction.

What is the reward of digital suction-effective mandibular complete denture manufacturing?

That is total patient satisfaction. That goose bump feel when you hear that popping sound in three stages, precision impression, digital denture try-in, and final digital denture delivery. We don't have to give the "lower denture speech"... the one designed to lower their expectations.

Precision impression



Fig 8-14 Do-able impression technique for everyone.





Fig 8-15 SEMCD not depending upon years of clinical experience.

Digital Denture try-in





Fig 8-16



Fig 8-17 Confirmation with suction action of the final lower denture.



Fig 8-18

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Biography of Dr. Jiro Abe



- Dr. Jiro Abe graduated from Tokyo Dental College in1981. He developed the suction mechanism of mandibular complete dentures in 1999 and published, "Successful Mandibular Complete Denture Suction for everyone". He has been diffusing it throughout the world since 2004.
- He was the director of the Academy of Clinical Dentistry from 1999 to 2005 and its councilor from 2005 to 2009.
- He founded the Japan Denture Association and has been its chairman from 2006 to 2015.
- He has stayed active as a former president of the Japan Plate Denture Association since 2015.
- He is the instructor of Ivoclar Vivadent BPS International Clinical and GC & MORITA Complete Denture seminars, the lecturer of the Japan Dental Association.
- He has been in various activities as a professor at the Tohoku University Graduate school of Dentistry since 2010 and at Kanagawa Dental College since 2012.
- Dr. Abe has also held many distinguished positions throughout the years and published Quintessence book. "4 Steps from Start to Finish Mandibular Suction Denture and BPS: a Perfect Manual-for All Types of Fully Edentulous Cases" in 2012 and "Mandibular Suction Effective Denture,The professional: Clinical and laboratory technique Class Ⅰ, Ⅱ, Ⅲ - with Aesthetics" in 2018, they were translated into English, Chinese and Korean.
- He has played active as an international committee member of the American Prosthodontic Society since 2015.

Contact Details

Dr. Jiro Abe

Abe Dental Clinic

1-12-43-2F Sengawa-cho Chofu-city, Tokyo Japan

Postal code 182-0002

Phone : +81-3-3300-1184

E-mail : jiroabe@ra2.so-net.ne.jp

YouTube : http://www.youtube.com/watch?v=rgsj5xAwSA8

Home Page : http://suctiondenture.jp/

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INTERACTION Co., Ltd.
2-13-1-202 Kyonan-cho, Musashino-city, Tokyo 180-0023, JAPAN
Tel : +81-70-6563-4151
Fax : +81-42-290-2927
URL : http://interaction.jp


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